

MARK GERGES

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Summary

Staff Software Engineer and technical lead on Tesla's internal Generative AI platform: the multi-provider LLM gateway, on-prem GPU fleet, access-control plane, and cost-attribution pipeline that teams across the company build on, including the systems serving external customers on Tesla.com. I came to infrastructure through hardware: three years building computer vision and opto-electro-mechanical inspection systems, and six years on cost, schedule, and risk for Gulf oil and gas capital projects before that. Throughout Tesla I go by "The Janitor," earned cleaning up messes and making sure everything works properly and reliably while delivering critical solutions. I would rather bring a measurement than an opinion.

Experience

Tesla Inc

Aug 2021 – Present

AI Platform · Remote

Nov 2024 – Present

Staff Software Engineer since Aug 2026; Sr. Software Engineer Nov 2024 – Aug 2026.

Technical lead since Mar 2025 for the platform serving GenAI models company-wide across billions of requests, at 55× year-over-year growth in users and roughly 6× growth in GPU capacity across three datacenters. Set technical direction across the gateway, metering, and serving layers, built the capacity and spend case that funded the growth, and manage the product team whose direction sets engineering priorities across the platform.

- Sole author of the inference benchmarking harness the platform's capacity decisions run on: concurrency stress testing at variable load, throughput and context-window saturation, and tail latency. Its time-to-first-token data was the empirical case behind the capacity proposals that grew the on-prem GPU fleet roughly 6× and put frontier open-weight models in house.
- Built the tenancy cost model behind capacity planning, resolving depreciation, power, and maintenance-headcount inputs from infrastructure and datacenter owners into a cost-per-token rate benchmarked against managed cloud, now the shared basis on which teams argue build-versus-rent.
- Built the provider- and API-agnostic inference gateway covering chat, embeddings, and image and video generation, driving the interface contract across backend, infrastructure, and model-provider teams so Claude Code, Codex, OpenCode, Grok CLI and others work against one endpoint for all Tesla employees.
- Own the metering and access-governance layer end to end: versioned rate cards, per-org cost attribution, automated spend limits, and manager-chain approval. Specified the attribution schema with the database team and the self-serve console with frontend, replacing a manual approval queue with a token service that took the platform out of the critical path for every new consumer.
- Run model onboarding across cloud and on-prem providers, taking frontier models from release to company-wide availability in days.
- Manage the product team, setting the direction and product decisions that drive priorities for the backend, frontend, and ML engineers building on the platform: the customer-facing assistant on Tesla.com and mobile, document extraction across millions of documents, social and manufacturing analytics assistants, and internal HR support that cut ticket volume nearly in half.

Sr. Mechatronics Engineer · San Diego, CA

Aug 2022 – Oct 2024

Designed, programmed, and constructed computer vision and opto-electro-mechanical inspection systems for manufacturing, from optics and hardware through model training and production deployment.

- Optimized and standardized optical and photometric parameter tuning to detect micron-sized defects, then trained and validated the object detection, classification, and segmentation models built on top.
- Designed and tested camera and scanner hardware, software, and inspection methodologies, extracting depth and dimensional data with open-source and custom algorithms.
- Iterated hardware and software against changing inspection needs until the designs were adopted globally, becoming the standard other teams and suppliers built to.

Sr. Technical Project Manager · Fremont, CA

Aug 2021 – Jul 2022

Managed high-visibility, high-risk vision and inspection programs from requirements gathering through global deployment.

- Integrated machine learning models and new hardware into production inspection systems with cross-functional teams, lifting inspection rates 30% and expanding hardware capability 700%.
- Wrote the documentation, procedures, and global training for vision technicians covering procurement, installation, programming, station design, and line integration, improving takt times while reducing safety incidents.
- Coordinated stakeholders, resources, and suppliers across programs, building supplier relationships that secured continuous access to new technology.

University of Washington

Feb 2020 – Jun 2021

Graduate Research Assistant, Optimal Lighting for Machine Vision · Seattle, WA

Feb 2020 – Jun 2021

- Calculated, simulated, and implemented optimal lighting distribution and placement using Revit, Simulink, MATLAB, and C, then programmed and constructed the control system.

- Eliminated glare and gradient on uneven topographies, raising defect identification from 94% to 96% while cutting false positives 1% and false negatives 2%. Resulted in a granted US patent, a journal publication, and a conference presentation.

Earlier Experience: Project Controls & Construction

Hamad Sulaiman Al-Ghanim Co.

Sep 2016 – Aug 2019

Project Control Engineer · Kuwait

Sep 2016 – Aug 2019

- Owned schedule, cost, and risk across the firm's portfolio: resource-loaded quantities, manhours, and costs on Primavera P6, built the cost-code system, and ran the weekly and monthly progress, forecast, and risk cycle.
- Attained ISO and OHSAS certification for the firm as management representative for quality assurance, auditing departments semi-annually for continued adherence.

Consolidated Contractors Co. (CC-EPC)

Dec 2013 – Aug 2016

Planning, Cost and Risk Engineer · Saudi Arabia

Nov 2014 – Aug 2016

- Scheduled and resource-loaded large capital projects on Primavera P6, revising forecast and budget costs against trends, claims, and BOQs.
- Led weekly progress meetings and monthly cost and risk reviews, wrote contractual correspondence to clients and subcontractors, and reported progress and forecast cost to headquarters.
- As Mechanical Site Engineer (Dec 2013 – Oct 2014), ran execution and testing of a large underground piping package (RTR and carbon steel), leading three foremen and 60 laborers plus the fabrication shop.

Selected Projects

Surface Defect Detection of Large Cast Bodies, Tesla, CA

Jan 2022 – Feb 2025

- Capitalized on surface texture and directional hypersensitivity; designed, constructed, programmed and deployed an in-motion 41-camera Multiview station and dual robotic-arm articulated inspection stations, adopted by other internal teams and suppliers.
- Coordinated delivery across seven functions: defect criteria with quality and materials science, cell layout and fixturing with mechanical design and construction, camera networking with networking, line integration with production, and models with machine learning.

Vehicle Status Inspection Station, Tesla, CA

Nov 2021 – Nov 2023

- Capitalized on network camera capabilities for remote calibration and photometric reflection to increase defect contrast; designed the optics, photometrics, and structure, delivering \$3M per quarter in cost recovery that grew with iteration.
- Coordinated the build across six groups: permitting and siting with the City of Fremont, structure and installation with construction, camera and lighting hardware with suppliers, vehicle routing with traffic, line integration with production, and models with machine learning.

Vehicle Service Triage Station, Tesla, CA

Jul 2024 – Jan 2025

- Designed the optics, photometrics, and structure of a service inspection station, cutting footprint and cost while increasing coverage for millimetre-accuracy defect and debris detection.

Technical Skills

- **Generative AI & LLM Infrastructure:** multi-provider gateways, on-prem inference serving (vLLM, KServe), inference benchmarking and load testing, model onboarding and evaluation, embeddings and reranking, GPU capacity planning
- **Infrastructure & Data:** Kubernetes, Helm, Envoy, Kafka, Redis, Grafana, PostgreSQL, ClickHouse, AMD ROCm, secure on-prem GPU infrastructure, metering pipelines
- **Providers & Programming:** Google Vertex AI, AWS Bedrock, Azure OpenAI, Anthropic, xAI; Python, Go, FastAPI, C, MATLAB
- **Computer Vision & Control:** object detection, classification and segmentation, optical and photometric design, SWIR and visible-light imaging, camera interfacing, embedded lighting and motion control, robotic-arm cells; OpenCV, YOLO, Simulink
- **Delivery & Leadership:** product team management, technical direction across backend, frontend and ML, cross-functional program delivery, capacity planning and FinOps, vendor and supplier management, ISO/OHSAS quality systems; Jira, Confluence, Primavera P6

Education

Master of Science, Mechanical Engineering, Controls and Robotics, University of Washington, Seattle, WA

2019 – 2021

- Thesis: adaptive lighting control for uniform and uneven topographies. [UW Research Archive](#)

Bachelor of Science, Mechanical Engineering, Mechatronics and Design, The American University in Cairo, Egypt

2008 – 2013

- Coursework: Machine Learning Specialization (Stanford Online); Computer Vision (Carnegie Mellon SCS); Data Structures and Algorithms, Python, Object-Oriented Java (Georgia Tech).

Publications, Patents & Talks

- US Patent [12,610,446](#), "Adaptive Illuminance Control," granted Apr 2026 (filed Apr 2022). Inventors: Mark Gerges, Xu Chen. Assignee: University of Washington.
- "Shape-Adaptive Lighting for Uneven and Non-Uniform Topographies," IEEE/ASME Trans. Mechatronics. Presented at MECC 2022.

Fields of Interest

GenAI Infrastructure • LLM Gateways • On-Prem GPU Serving • Platform Economics • Distributed Systems • Computer Vision